

45 Days Summer Training Program

Assignment: 3

1. Create a class named Student with attributes:

- o name

- o age

- o course

Create an object and print all the details.

```
➤ class Student:
```

```
    def __init__(self, name, age, course):
```

```
        self.name = name
```

```
        self.age = age
```

```
        self.course = course
```

```
student1 = Student("Pankaj", 20, "Computer  
Science")
```

```
print("Name:", student1.name)
```

```
print("Age:", student1.age)
```

```
print("Course:", student1.course)
```

- output :-

Name: Pankaj

Age: 20

Course: Computer Science

2. Create a class named Car with attributes:

- o brand

- o model

- o price

Create two objects and display their details.

```
➤ class Car:
```

```
    def __init__(self, brand, model, price):  
        self.brand = brand  
        self.model = model  
        self.price = price
```

```
car1 = Car("Toyota", "Corolla", 1500000)
```

```
car2 = Car("Honda", "City", 1800000)
```

```
print("Car 1")
```

```
print("Brand:", car1.brand)
```

```
print("Model:", car1.model)
```

```
print("Price:", car1.price)
```

```
print("\nCar 2")
```

```
print("Brand:", car2.brand)
```

```
print("Model:", car2.model)
```

```
print("Price:", car2.price)
```

- output :-

Car 1

Brand: Toyota

Model: Corolla

Price: 1500000

Car 2

Brand: Honda

Model: City

Price: 1800000

3. Create a class named Employee and use a constructor (`__init__`) to initialize:

- o employee_id

- o name

- o salary Print the employee details.

➤ class Employee:

```
def __init__(self, employee_id, name, salary):
```

```
    self.employee_id = employee_id
```

```
    self.name = name
```

```
    self.salary = salary
```

```
emp = Employee(101, "Pankaj", 50000)
```

```
print("Employee ID:", emp.employee_id)
```

```
print("Name:", emp.name)
```

```
print("Salary:", emp.salary)
```

- output :-

Employee ID: 101

Name: Pankaj

Salary: 50000

4. Create a class named Rectangle with attributes:

o length

o width

Create a method to calculate and print the area of the rectangle.

➤ class Rectangle:

```
def __init__(self, length, width):
```

```
    self.length = length
```

```
    self.width = width
```

```
def area(self):
```

```
    print("Area of Rectangle:", self.length *  
self.width)
```

```
rect = Rectangle(10, 5)
```

```
rect.area()
```

- output :-

Area of Rectangle: 50

5. Create a class named Circle with an attribute radius.

Create a method that calculates and prints the area of the circle

```
➤ class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        print("Area of Circle:", 3.14 * self.radius *
self.radius)

c = Circle(7)
c.area()
```

- output :-

Area of Circle: 153.86

6. Create a class named BankAccount with:
 - o account_holder
 - o balance Create methods:
 - o deposit(amount)
 - o withdraw(amount) Display the updated balance after each transaction.

```
➤ class BankAccount:
    def __init__(self, account_holder, balance):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
```

```
self.balance += amount  
print("Balance after deposit:", self.balance)
```

```
def withdraw(self, amount):  
    self.balance -= amount  
    print("Balance after withdrawal:",  
self.balance)
```

```
acc = BankAccount("Pankaj", 10000)
```

```
acc.deposit(2000)  
acc.withdraw(1500)
```

- output :-

Balance after deposit: 12000

Balance after withdrawal: 10500

7. . Create a class named Animal with a method sound(). Create a child class Dog that overrides the sound() method and prints "Bark".

```
➤ class Animal:  
    def sound(self):  
        print("Animal Sound")
```

```
class Dog(Animal):  
    def sound(self):  
        print("Bark")
```

```
d = Dog()
d.sound()
```

- output :-

Bark

8. Create a class named Person with attributes:

o name

o age Create a child class Student with an additional attribute:

o roll_number

Display all the details using an object of the Student class.

➤ class Person:

```
def __init__(self, name, age):
    self.name = name
    self.age = age
```

class Student(Person):

```
def __init__(self, name, age, roll_number):
    super().__init__(name, age)
    self.roll_number = roll_number
```

```
s = Student("Pankaj", 20, 101)
```

```
print("Name:", s.name)
```

```
print("Age:", s.age)
```

```
print("Roll Number:", s.roll_number)
```

- output :-

Name: Pankaj

Age: 20

Roll Number: 101

9. Create a class named Calculator with methods:

o add()

o subtract()

o multiply()

o divide()

Take two numbers as input and perform all operations.

➤ class Calculator:

def add(self, a, b):

print("Addition:", a + b)

def subtract(self, a, b):

print("Subtraction:", a - b)

def multiply(self, a, b):

print("Multiplication:", a * b)

def divide(self, a, b):

print("Division:", a / b)

a = int(input("Enter first number: "))

b = int(input("Enter second number: "))


```
c = Calculator()
```

```
c.add(a, b)
```

```
c.subtract(a, b)
```

```
c.multiply(a, b)
```

```
c.divide(a, b)
```

- output :-

Enter first number: 10

Enter second number: 5

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2.0

10. Create a class named LibraryBook with attributes:

- book_name
- author
- price

Create a method display_book_info() that prints all book details.

Create at least three book objects and display their information.

➤ class LibraryBook:

```
def __init__(self, book_name, author, price):
```

```
    self.book_name = book_name
```

```
    self.author = author
```

```
    self.price = price
```

```
def display_book_info(self):  
    print("Book Name:", self.book_name)  
    print("Author:", self.author)  
    print("Price:", self.price)  
    print()
```

```
book1 = LibraryBook("Python Basics", "John  
Smith", 450)  
book2 = LibraryBook("Data Structures", "Ravi  
Kumar", 550)  
book3 = LibraryBook("Machine Learning",  
"Andrew Ng", 750)
```

```
book1.display_book_info()  
book2.display_book_info()  
book3.display_book_info()
```

- output :-

Book Name: Python Basics
Author: John Smith
Price: 450

Book Name: Data Structures
Author: Ravi Kumar
Price: 550

Book Name: Machine Learning
Author: Andrew Ng

Price: 750